PyGame Car Crash Tutorial - Part 7

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Time required: 30 minutes

Let's finish up our car game with sounds, fonts, and a scoring system.

config.py

```python
"""
    Filename: config.py
    Author:
    Date:
    Purpose: Global variables and constants for the entire program
"""
# Import config module into all other modules

# Setup global constants and variables for screen size
WIDTH = 400
HEIGHT = 600

# Global variable for speed across screen
SPEED = 4

# Constant for how much the speed increases each enemy car pass
SPEED_INCREASE = .4

BLACK = (0, 0, 0)
```

enemy.py

To keep score, initialize a score variable in the enemy.py class.

```python
class Enemy(pygame.sprite.Sprite):
    """Define the enemy class and methods"""
# ------------------------- INITIALIZE ENEMY OBJECT ------------------------#

    def __init__(self):
        self.score = 0
        """Construct an enemy object from Sprite class"""

        # Call the constructor of the superclass (pygame.sprite.Sprite)
        super().__init__()

        self.score = 0
        self.speed = config.SPEED

        # Load enemy car image from file into a variable
        self.image = pygame.image.load(
            "./assets/enemy.png").convert_alpha()
```

We keep track of how many times the Enemy gets to the bottom of the screen. Each time this happens, the player gets one more point.

```python
    # -------------------------- UPDATE ----------------------------------#
    def update(self):
        # Move the sprite down SPEED pixels at a time
        self.rect.move_ip(0, self.speed)

        # When the top of the sprite reaches the botton of the surface
        if (self.rect.top > config.HEIGHT):
            # Get a random location 40 pixels away from left and right.
            x = randint(40, config.WIDTH - 40)

            # Move car above the program window
            y = -120

            # Move car to beginning position
            self.rect.center = (x, y)

            # Increase speed each time the enemy car starts at the top
            self.speed += config.SPEED_INCREASE

            # Increment score every time the player dodges an oncoming car
            self.score += 1
```

car_crash_7.py

We are going to add another library to display our game over screen.

```
pip install pygame-menu
```

```python
"""
    Filename: car_crash_7.py
    Author:
    Date:
    Purpose: Add scoring, sound and a game over menu
"""
# pip install pygame-ce
# Import modules
import pygame
# pip install pygame-menu
import pygame_menu as pm
from sys import exit
from time import sleep
# Import our game classes
import config
import player
import enemy
```

Sound

```python
class CarCrash:
    def __init__(self):
        # Pre initalize mixer with larger buffer size for better performance
        pygame.mixer.pre_init(
            44100,  # frequency (Hz)
            16,     # bit depth
            2,      # number of channels, 1 mono, 2 stereo
            4096    # buffer size, larger to optimize music playback.
        )

        # Initialize the pygame library
        pygame.init()
```

```python
        # Optimize game by only allowing these events to be captured
        pygame.event.set_allowed(
            [pygame.QUIT, pygame.KEYDOWN, pygame.KEYUP]
        )

        # Set window icon
        window_icon = pygame.image.load('./assets/car.ico')
        pygame.display.set_icon(window_icon)

        # Create the player and enemy sprites
        self.create_sprites()

        # Create System font object for score
        self.font_small = pygame.font.SysFont("arialblack", 20)

        # Load sound file into memory
        pygame.mixer.music.load('./assets/background_music.wav')

        # Set volume to 30%, range from 0.0 (mute) to 1.0 (full volume)
        pygame.mixer.music.set_volume(0.3)

        # Play in a loop until stopped
        pygame.mixer.music.play(-1)
```

Add the display_game_over_method.

There are different themes you can choose for the game_over object. This example uses THEME_BLUE. You can use any of the following to customize your menu.

```
THEME_BLUE, THEME_DARK,THEME_DEFAULT, THEME_GREEN, THEME_ORANGE,
THEME_SOLARIZED'
```

```python
# ------------------------- DISPLAY GAME OVER ----------------------------#
    def display_game_over(self):
        """Display game over on top of the stopped game"""
        # Stop background sound
        pygame.mixer.music.stop()

        # Play crash sound
        crash = pygame.mixer.Sound("assets/crash.wav")
        crash.play()
        crash.set_volume(0.5)

        # Wait 1 second
        sleep(1)

        # Define a menu object for the game over screen
        game_over = pm.Menu(
            title="Game over",      # Set title menu to "Game over"
            width=config.WIDTH,     # Set to width of game surface
            height=config.HEIGHT,   # Set to height of game surface
            # Set the theme of the menu to an orange color scheme
            theme=pm.themes.THEME_ORANGE
        )

        # Display final score
        game_over.add.label(f"Score: {self.enemy_sprite.score}")

        # Add label to provide space between buttons
        game_over.add.label("")

        # Add a button to the game over menu for exiting the game
        game_over.add.button(
            title="Play Again?",    # Button text
            action=main             # Call main() to start over
        )

        # Add label to provide space between buttons
        game_over.add.label("")

        # Add a button to the game over menu for exiting the game
        game_over.add.button(
            title="Exit",           # Button text
            action=pm.events.EXIT   # Exit the game when clicked
        )

        # Run the main loop of the game over menu on the specified surface
        game_over.mainloop(self.surface)
```

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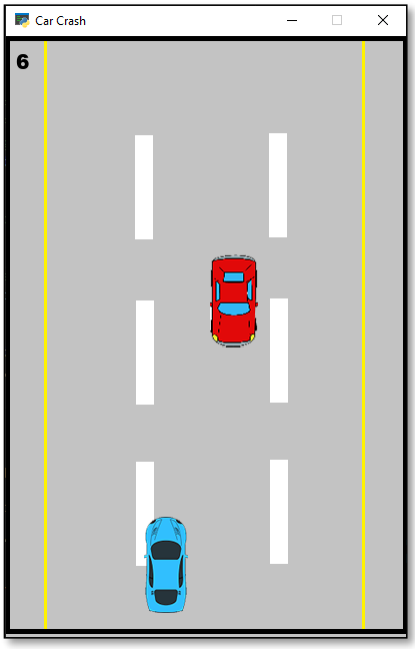
```
THEME_BLUE, THEME_DARK,THEME_DEFAULT, THEME_GREEN, THEME_ORANGE,
THEME_SOLARIZED'
```

```python
# ---------------------- GAME LOOP -------------------------------------#
    def game_loop(self):
        """Start the infinite Game Loop"""
        while True:
            self.check_events()
            self.check_collision()

            # --------------- DRAW ON BACKBUFFER ------------------------#
            # Draw everything on the backbuffer first
            # Fill the surface with the background image loaded earlier
            self.surface.blit(self.background, (0, 0))

            # ------------- UPDATE AND DRAW SPRITES ---------------------#
            # Run the update method on all sprites
            self.all_sprites.update()

            # Draw all sprites on the surface
            self.all_sprites.draw(self.surface)

            # Render score before drawing it on the surface
            self.score = self.font_small.render(
                str(self.enemy_sprite.score), True, config.BLACK
            )

            # Draw score on the surface
            self.surface.blit(self.score, (10, 10))

            # --------------- UPDATE SURFACE ----------------------------#
            # From backbuffer, update Pygame display to reflect any changes
            pygame.display.update()

            # Cap game speed at 60 frames per second
            self.clock.tick(60)


def main():
    # Create game instance
    car_crash = CarCrash()
    # Start the game
    car_crash.game_loop()


main()
```

Example run:



What's Next?

There is much more that can be done with this game. Here are some ideas for you to practice and implement on your own.

- Keep track of the score between games.
- Multiple enemies spawning after set periods of time. (Similar to how we increased speed after a set period of time)
- Add some additional audio to the game, such as movement sounds (audio that plays when you move the character)
- Add the concept of multiple Lives or a Health bar.
- Variations in the shape and size of the "enemies".

Assignment Submission

Zip up the program files folder and submit in Blackboard.